Mauna Loa CO2 Data Time-Series Analysis

STAT4601 Time Series Analysis

Department of Statistics & Actuarial Science

Faculty of Science

The University of Hong Kong

Wong Chi Ho 3036309347

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1. Introduction and Data Collection

1.1 Project Overview

Atmospheric carbon dioxide (CO_2) concentrations represent one of the most critical environmental indicators of climate change. Leads to profound implications for global warming, ocean acidification, and ecosystem stability. The continuous monitoring of CO2 levels in Mauna Loa Observatory in Hawaii has provided the longest uninterrupted record of atmospheric CO2 measurements since 1958 (1). This dataset demonstrates a persistent upward trend and seasonal patterns driven by different environmental and anthropogenic factors.

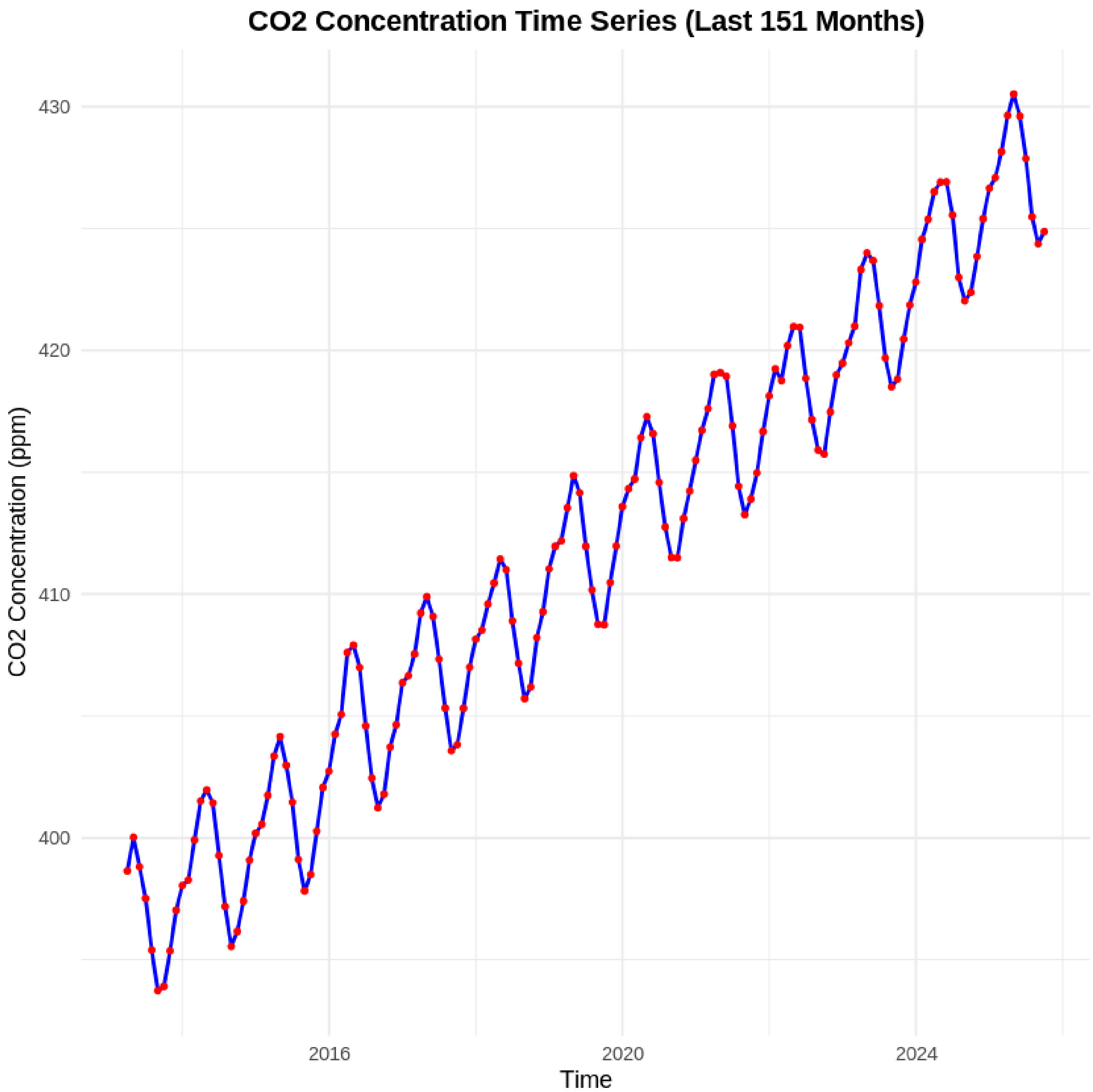

Figure 1.1: NOAA Global Monitoring Laboratory monthly mean carbon dioxide concentration (04/2013 - 10/2025)

This report employs Seasonal Autoregressive Integrated Moving Average(SARIMA) modeling to analyze and forecast the atmospheric CO2 concentrations globally. It uses Box-Jenkins methodology to identify models, estimate parameters, and do diagnostic checks to ensure robust model selection and validation. Predictions would be made after selecting the most suitable model.

2. Data Stationarity

2.1 Initial Data Examination and Visual Assessment

Through direct observation, we find that the temperature is not stationary throughout the years preliminarily. Since stationarity represents a fundamental assumption underlying most time series modeling approaches, the visual inspection of the original time series plot reveals an upward trend in data, providing a great hint in non-stationary characteristics. We can also observe regular periodic oscillations with approximately 12 month cycles. To quantify the visual observations, we employ Augmented Dickey-Fuller (ADF) tests with lag 13 to give a more precise conclusion.

Times of log	p-value (ADF)
0	7.14e-01
1	6.18e-01
2	6.07e-01

Table 2.1: ADF test result after multiple log

The ADF test on the original CO2 data produced a p-value of 7.14e-01. With the p-value exceeding 0.05 significance threshold, we fail to reject the null hypothesis. We can conclude the original atmospheric CO2 concentration time series is non-stationary.

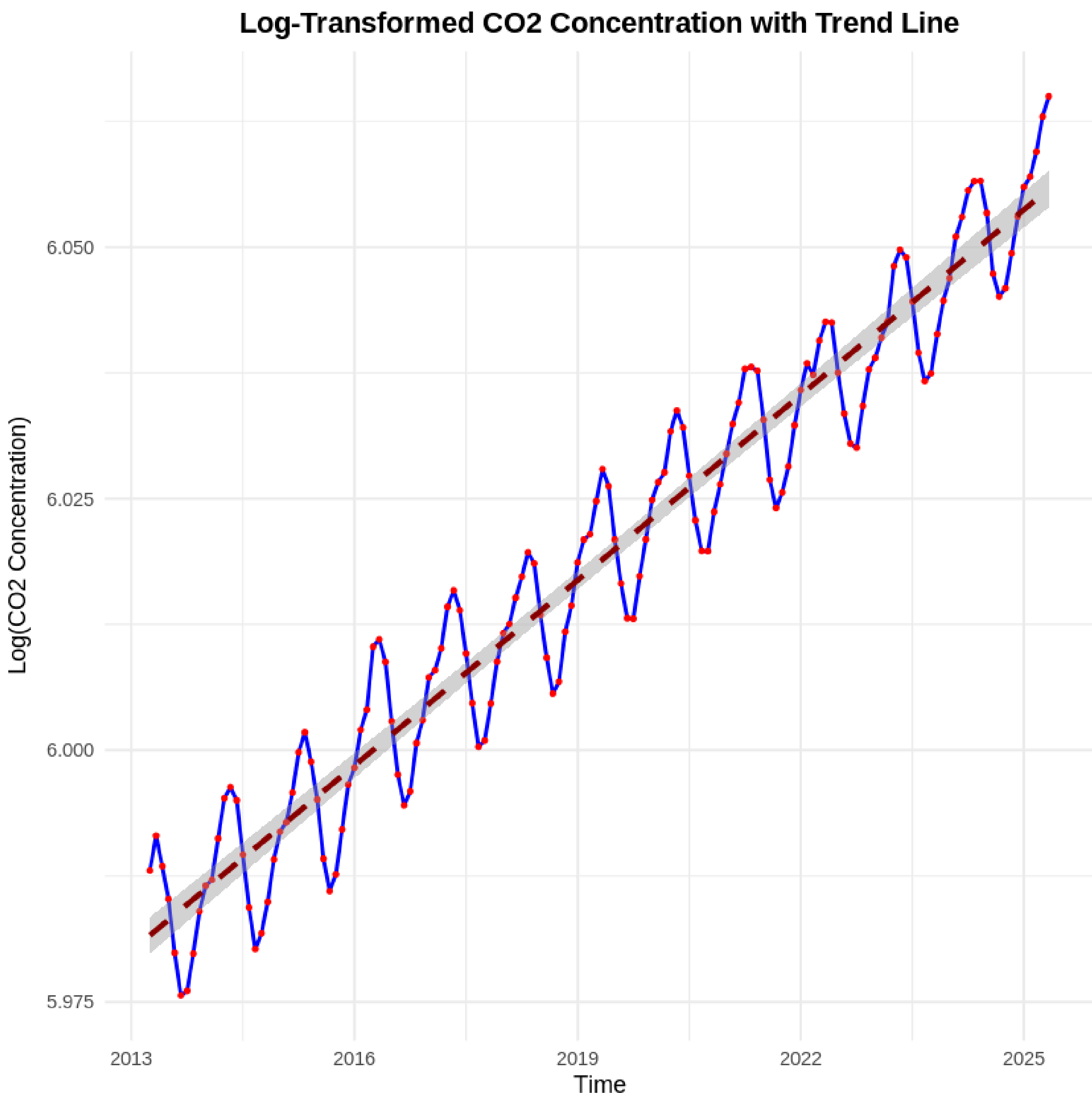

Figure 2.2 Time series after taking log

The trend line obviously exists. This indicates differencing is needed and the log did not do much transformation on the time series data. We would conduct first differencing: $\delta Z_t = Z_t - Z_{t-1}$. The time series after first differencing is shown below.

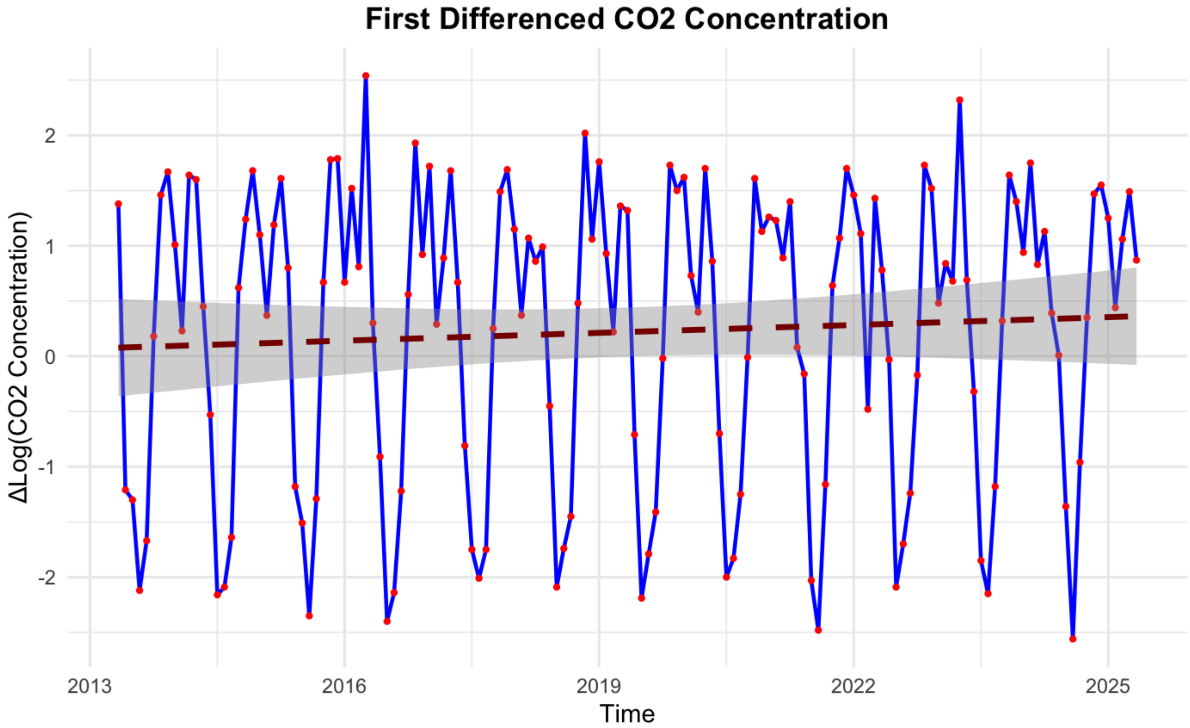

Figure 2.3: Time series after first difference

Stationary is reached for our CO2 concentration data. We further test to see if the mean and variance is constant. We divide the data into three segments with 36 data points each and use ANOVA test to see if there are any statistically significant differences between overall mean, following with Bartleet's test on testing the variance difference.

Segment	Time Period	Mean	Variance
1	1-36	0.2489	2.0537
2	37-72	0.1650	1.8582
3	73-108	0.1847	1.8301
4	109-144	0.2625	1.6851

Table 2.1: Mean and Variance of difference segment on CO2 time series data

The first-differenced CO_2 series demonstrates weak stationarity by both visual and statistical evidence. The ANOVA test (p-value = 0.9876 > 0.05) and Bartlett's test (p-value = 0.9349 > 0.05) provide strong statistical support for constant mean and variance across temporal segments.

We further test if the model is white noise using the Ljung-Box test(lag 20). The result is p-value(0.00e+00) less than 0.05 with significant signals from lag 1 to lag 20 on the ACF plot (Fig. 3.1.1). Hence the time series is not white noise. We can conduct model building.

3. Model Analysis

3.1 Seasonal Autoregressive Integrated Moving Average (SARIMA) Model Detection

By observing the ACF and PACF, we can identify a preliminary time series model based on the number of lag k that exceeds the boundary of Bartlett' s approximation.

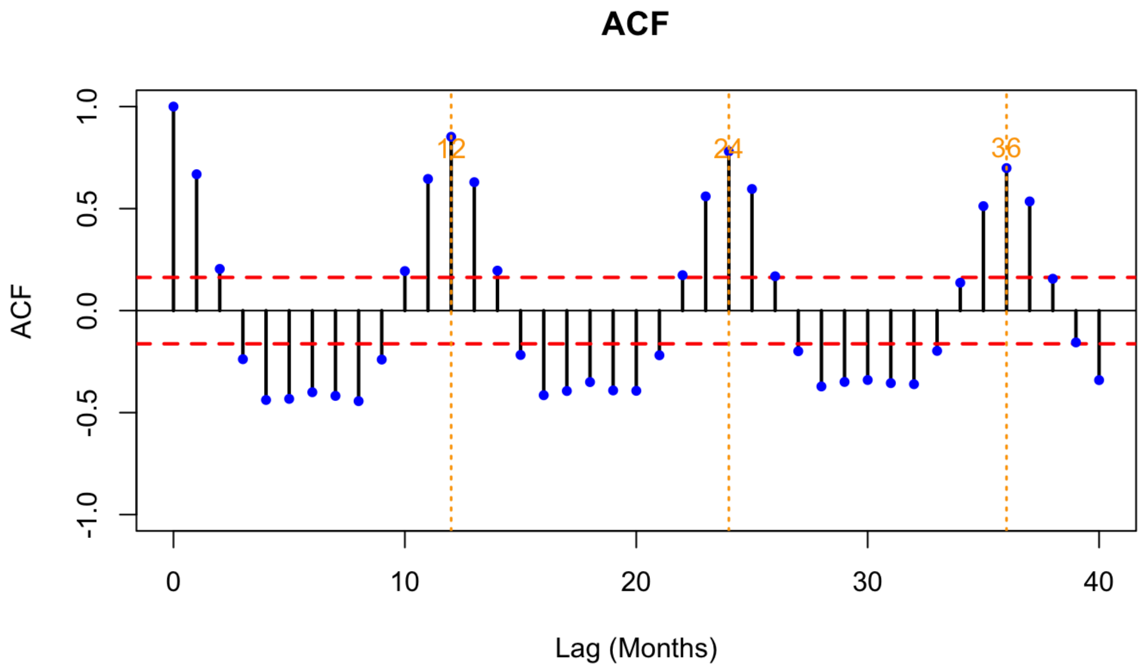

Figure 3.1.1: ACF after first log and first difference

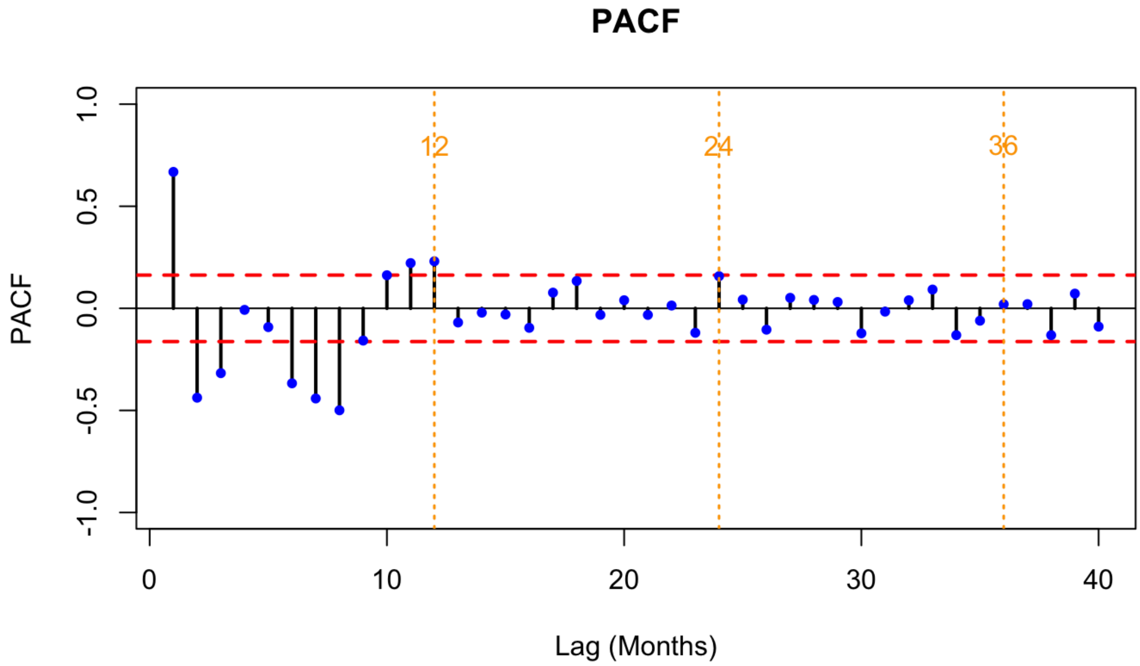

Figure 3.1.2: PACF after first log and first difference

In figure 3.1, we can see the ACF decay slowly from lag 1 to lag 11. In figure 3.2, there are significant spikes at lags 1 (+0.668), 2 (-0.438), 3 (-0.238) to lags 33(-0.197). Although most lags exceed Bartlett's approximation boundary which means most of them are significant in ACF, to be cautious, we choose ARIMA(1,1,0), the simpler model at first as we could add more parameters in the over-paramatization section. The PACF have cut off at lag 3. Although lag 6,7,8 exceeds Bartlett's approximation boundary, indicating they are also significant, we still choose to use ARIMA(0,1,3) for the same reason.

For the seasonal component, the ACF graph has massive spikes at lags 12 (0.852), 24(0.780), 36(0.699), showing a 12-month seasonality. There are also significant spikes at lag 12 in PACF, but the neighbouring lags 11 and 13 do not have the same magnitude, one positive and one negative, and they are cut off at lag 12.

Therefore, we suggest ARIMA(1,1,0)(0,1,1)[12] or ARIMA(0,1,3)(0,1,1)[12]. We continue to do residual analysis.

3.2 Model diagnostics

After going through initial model specification, we conduct the Shapiro-Wilk normality test, plot histogram with Normal Density Function and Kernel Density Function(KDE), and Quantile-Quantile plot to observe the normality of the residual.

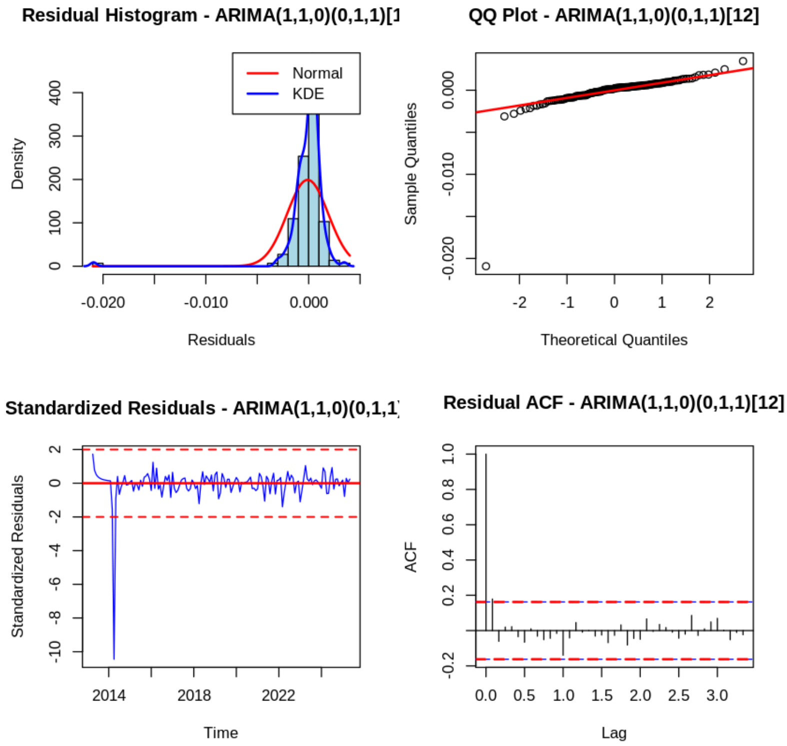

Figure 3.2.1: ARIMA(1,1,0)(0,1,1)[12] residual analysis

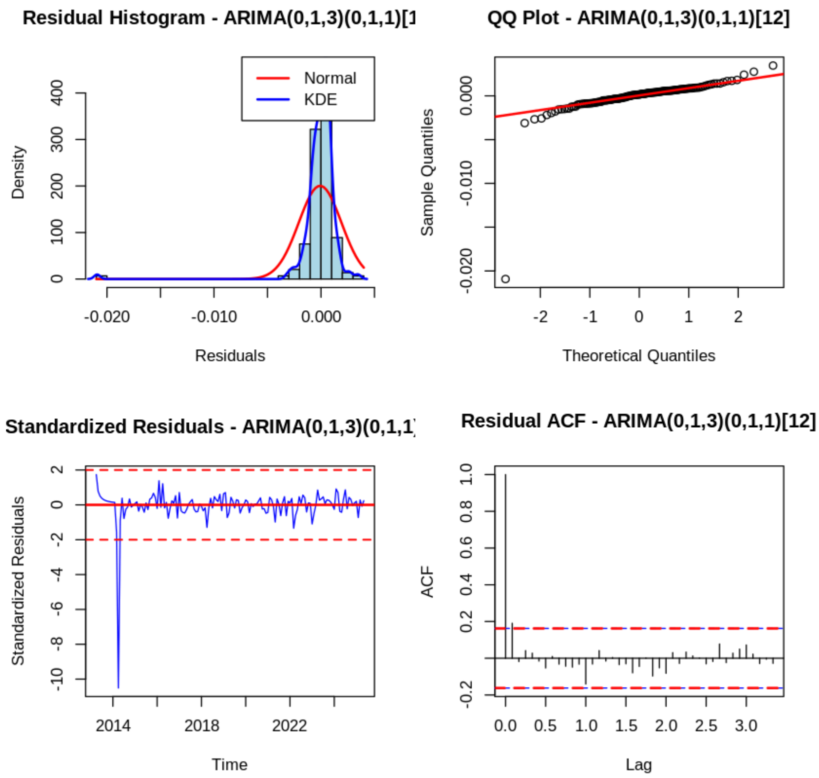

Figure 3.2.2: ARIMA(0,1,3)(0,1,1)[12] residual analysis

From figure 3.2.1 and 3.2.2 we can see both are fat tails on the left of the histogram and the KDE has a spike at 0, way higher than the normal density function. There is also a spike on the correlograms, indicating that there is remaining autocorrelation that the model did not capture. Besides, there are more than 6 data deviations from the normal redline for both models.

This fits the Shapiro-Wilk normality test result which both models have p-value lower than 0.0000, rejecting the null hypothesis that residuals follow normal distribution. Indicating the residuals are not normal.

Therefore, the residual sequence did not follow a normal distribution.

Next, we conduct Ljung-box test and over paramatize method

3.3 Ljung-Box test

Next, we perform a Ljung-Box test to see if these models are adequate.

Test Statistic: $$Q^* = n(n+2)\sum_{k=1}^{K}\frac{\hat{p}_k^2}{n-k}$$

Distribution: χ^2_{K-m}

The Ljung-Box Portmanteau test is used for validating the critical assumption of independent error terms. We reject the null hypothesis if the p-value is larger than 0.05 in the first 15 lags on the ACF.

```
Model 1 - ARIMA(1,1,0)(0,1,1)[12]:
Lag  P-value    Autocorrelation?
-----------------------------
 1   0.0290     ✗
 2   0.0700     ✓
 3   0.1460     ✓
 4   0.2438     ✓
 5   0.3431     ✓
 6   0.3900     ✓
 7   0.5030     ✓
 8   0.5943     ✓
 9   0.6485     ✓
10   0.7072     ✓
11   0.7800     ✓
12   0.5836     ✓
13   0.6404     ✓
14   0.6870     ✓
15   0.7525     ✓
```

Figure 3.3.1: Ljung Box test statistics for ARIMA(1,1,0)(0,1,1)[12]

```
Model 2 - ARIMA(0,1,3)(0,1,1)[12]:
Lag  P-value    Autocorrelation?
-----------------------------
 1   0.0201     ✗
 2   0.0655     ✓
 3   0.1265     ✓
 4   0.2124     ✓
 5   0.3197     ✓
 6   0.3906     ✓
 7   0.5038     ✓
 8   0.5945     ✓
 9   0.6611     ✓
10   0.7119     ✓
11   0.7735     ✓
12   0.5728     ✓
13   0.6399     ✓
14   0.6910     ✓
15   0.7544     ✓
```

Figure 3.3.2: Ljung Box test statistics for ARIMA(0,1,3)(0,1,1)[12]

For both models, they both have autocorrelation after lag 1 since both models have p-value larger than 0.05 after lag 1. This means we might need to add an extra parameter into those models. Therefore, we continue to conduct further testing.

3.4 Overparameterizing, Normality testing and AIC for model selection

In this section, we would conduct overparameterizing for ARIMA(1,1,0)(0,1,1)[12] and ARIMA(0,1,3)(0,1,1)[12] since the previous models are obviously not sufficient.

```
=== MODEL 1: ARIMA(1,1,0)(0,1,1)[12] ===

ARIMA(2,1,0)(0,1,1)[12]:
AIC: 135.45 (Original: 138.53)
AR2 coefficient: -0.1958 (p-value: NA)
Shapiro p-value: 0.0091

ARIMA(1,1,1)(0,1,1)[12]:
AIC: 131.89 (Original: 138.53)
MA1 coefficient: -0.6376 (p-value: NA)
Shapiro p-value: 0.0079

ARIMA(1,1,0)(1,1,1)[12]:
AIC: 139.26 (Original: 138.53)
SAR1 coefficient: -0.1562 (p-value: NA)
Shapiro p-value: 0.0118
```

Figure 3.4.1: Overparameterizing statistics for ARIMA(1,1,0)(0,1,1)[12]

```
=== MODEL 2: ARIMA(0,1,3)(0,1,1)[12] ===

ARIMA(1,1,3)(0,1,1)[12]:
AIC: 135.60 (Original: 155.96)
AR1 coefficient: -0.5037 (p-value: NA)
Shapiro p-value: 0.0077

ARIMA(0,1,4)(0,1,1)[12]:
AIC: 135.33 (Original: 155.96)
MA4 coefficient: 0.0733 (p-value: NA)
Shapiro p-value: 0.0115

ARIMA(0,1,3)(1,1,1)[12]:
AIC: 135.43 (Original: 155.96)
SAR1 coefficient: -0.0846 (p-value: NA)
Shapiro p-value: 0.0065
```

Figure 3.4.1: Overparameterizing statistics for ARIMA(0,1,3)(0,1,1)[12]

However, all the proposed new models have p-value less than 0.05 in the Shapiro-Wilk test, showing that all the overparameterized models' residuals do not follow normal. Since overparameterizing methods do not work, this might mean the initial estimated model is overparameterized. Therefore, we consider removing some parameters and do stepwise search on repeatedly doing model diagnostics, until we find a test that passes the Shapiro-Wilk test.

```
*** TOP MODELS (Sorted by AIC) ***
Model                        AIC     Shapiro p
----------------------------------------------
ARIMA(1,0,1)(0,1,1)[12]    138.25   0.0890
ARIMA(1,0,2)(0,1,1)[12]    138.44   0.0921
ARIMA(1,0,2)(0,1,2)[12]    140.07   0.0534
ARIMA(3,0,0)(0,1,1)[12]    142.04   0.0781
ARIMA(1,0,1)(1,1,2)[12]    142.46   0.0585
ARIMA(1,0,1)(0,1,3)[12]    142.52   0.0599
ARIMA(2,0,3)(0,1,1)[12]    142.85   0.0998
ARIMA(2,0,2)(0,1,1)[12]    142.90   0.1128
ARIMA(1,0,1)(2,1,2)[12]    142.93   0.0603
ARIMA(1,0,2)(0,1,3)[12]    143.61   0.0697
```

Through stepwise search, we found 72 models that pass the Shapiro-Wilk test. We then choose the model based on the model complexity and Akaike information criterion (AIC) to conduct testing on the Ljung-Box test as the data size is small at around 140 observations. By comparison, the ARIMA(1,0,1)(0,1,1)[12] is the simplest model with lowest AIC, therefore we conduct the residual testing on it.

3.5 Model Selection

After the model selection, we found that the ARIMA(1,0,1)(0,1,1)[12] successfully passed all the tests. Results are as follow:

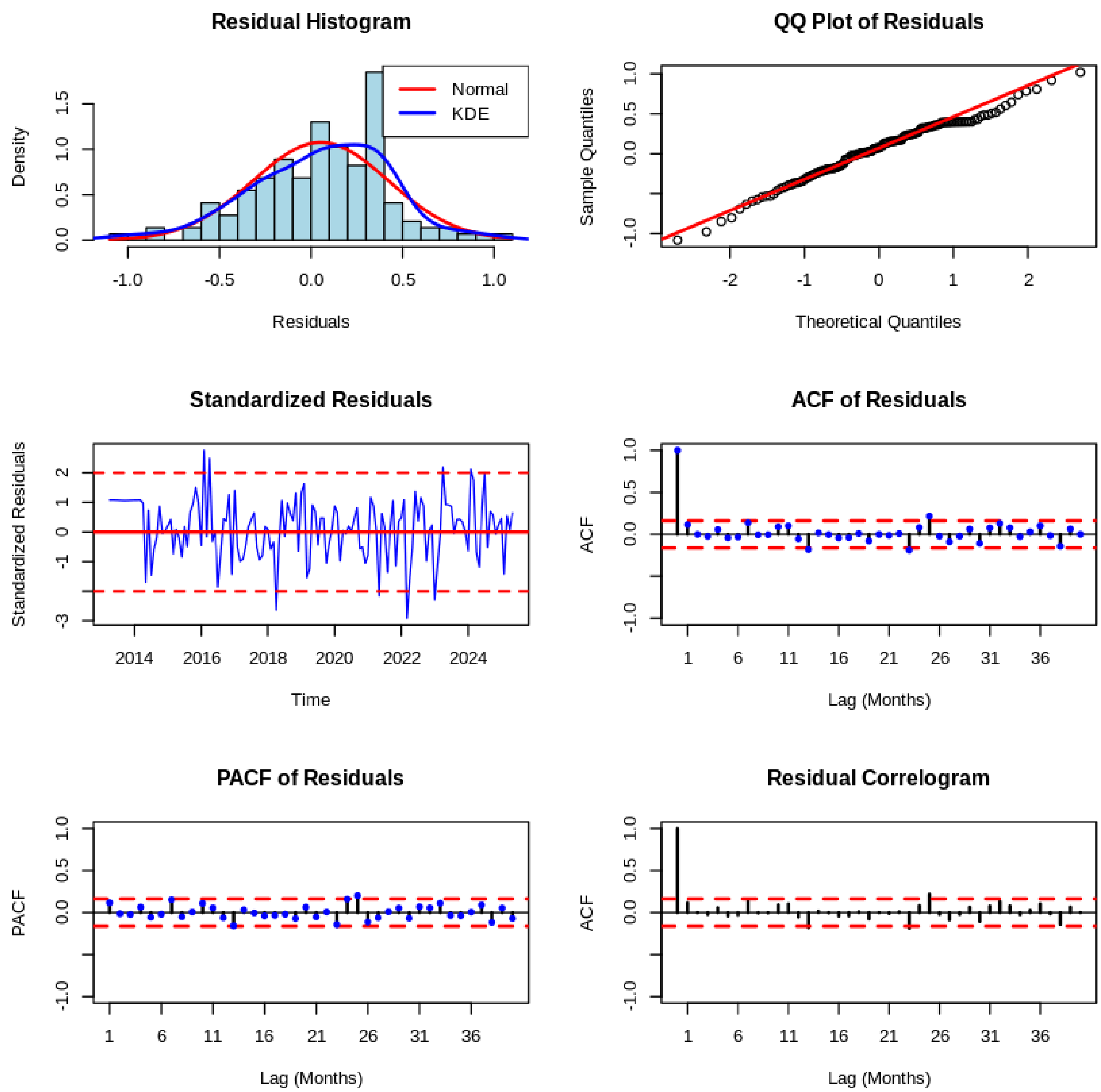

Figure 3.5.1: Result of ARIMA(1,0,1)(0,1,1)[12] residual test

Judging by the QQ plot, we can see the residuals are approximately normal as the points follow the straight line reasonably well, with only a little deviation at 1.5, this fits the observation on the histogram with bell shape distribution centered around 0 which the KDE deviated a bit. In the ACF, most of the lags are insignificant, no systematic pattern. For the PACF, all lags are insignificant, no AR structure remaining, showing the model captured all autoregressive components. Considering the Ljung-Box tests, all lags are significant, there is no autocorrelation between residuals, confirming all residuals are white noise, see figure 3.4.2.

```
*** LJUNG-BOX TESTS (Lags 1-15) ***
Lag  P-value    Autocorrelation?
----------------------------
 1   0.1567      ✗
 2   0.3667      ✗
 3   0.5516      ✗
 4   0.6260      ✗
 5   0.7203      ✗
 6   0.8050      ✗
 7   0.5365      ✗
 8   0.6435      ✗
 9   0.7362      ✗
10   0.6927      ✗
11   0.6212      ✗
12   0.6612      ✗
13   0.3205      ✗
14   0.3892      ✗
15   0.4629      ✗
```

Figure 3.4.2: ARIMA(1,0,1)(0,1,1)[12] Ljung-Box test result

3.6 Parameter Estimation

We estimated the parameters for the transformed series on ARIMA(1,0,1)(0,1,1)[12], the best model we selected for CO2 concentration data.

Model Equation:

$$(1-\phi_1 B)(1-B^{12})Z_t = (1-\theta_1 B)(1-\Theta_1 B^{12})a_t$$

Model Expand form:

$$Z_t = Z_{t-12} + \phi_1(Z_{t-1} - Z_{t-13}) + a_t - \theta_1 a_{t-1} - \Theta_1 a_{t-12} + \theta_1\Theta_1 a_{t-13}$$

Parameter Estimates (ML method):

- $\phi_1(\mathrm{AR1}) : 0.9999 \pm 0.0002$
- $\theta_1(\mathrm{MA1}) : -0.4640 \pm 0.0898$
- $\Theta_1(\mathrm{SMA1}) : -0.7842 \pm 0.0901$
- $\sigma^2(\text{Error variance}) : 0.1554$

Final Numerical Equation:

$$(1-0.9999B)(1-B^{12})Z_t = (1+0.4640B)(1+0.7842B^{12})a_t$$

Model Summary:

- Log Likelihood: -65.13

- AIC: 138.25
- AIC: 138.56
- BIC: 149.84

```
*** PARAMETER SIGNIFICANCE TESTS (Wald) ***
Parameter Estimate  Std.Err  z-value  p-value
ar1        0.9999   0.0002 4424.8158   0.0000 *
ma1       -0.4640   0.0898  -5.1659   0.0000 *
sma1      -0.7842   0.0901  -8.7018   0.0000 *
drift          NA       NA       NA      NANA
```

Figure 3.6.1: ARIMA(1,0,1)(0,1,1)[12] parameter significance test result

The figure above shows that the AR1,MA1,SMA1 are all significant using Wald test as all p-values less than 0.05.

4. Prediction

Based on the final numerical equation, we can calculate the parameter estimation on 5 data points.

Period	Actual	Predicted	Lower Bound	Upper Bound
1	429.61	430.0400	429.2670	430.8131
2	427.87	428.1959	427.3189	429.0729
3	425.48	426.0562	425.0863	427.0261
4	424.37	424.8244	423.7698	425.8790
5	424.87	425.0967	423.9638	426.2297

Table 4.1: Actual value and prediction of CO_2 concentration

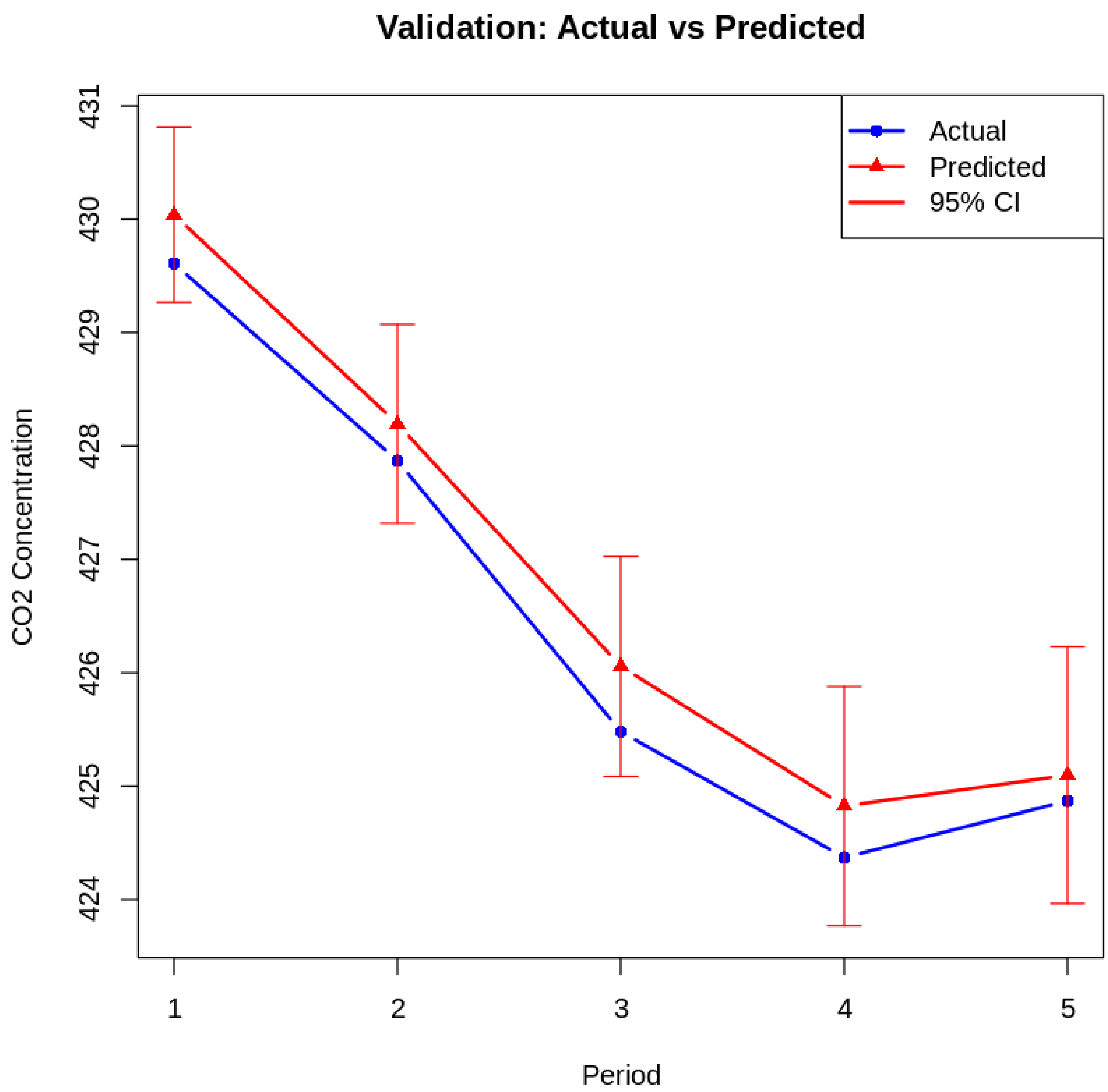

Figure 4.1: Comparison of the actual and prediction value of CO_2 concentration

The figure above visualized the prediction with 95% confidence interval. Since the actual data points fall in the 95% confidence interval of the prediction, we consider the prediction fairly good.

5. Conclusion

In this project, we identified and validated an optimal Seasonal ARIMA model for forecasting atmospheric CO2 concentrations from the Mauna Loa dataset using Box-Jenkins methodology. We start by identifying a preliminary time series model, then estimate the model parameters and do model diagnostic checking. Once we find the preliminary model is not adequate, we loop back to identify another time series model and continue to do the model diagnostic process, until we get the optimal model.

Through an iterative process of model specification, diagnostic checking, and stepwise refinement, the SARIMA(1,0,1)(0,1,1)[12] model was identified as the most appropriate. The model passed all diagnostic tests, including the Ljung-Box test and residuals normality tests with the lowest AIC, confirming that the residuals are white noise with no remaining autocorrelation and that the residuals are approximately normally distributed, and all estimated parameters were statistically significant.

The prediction performance of the model is robust, with observed values falling within the 95% prediction intervals. In summary, the SARIMA(1,0,1)(0,1,1)[12] model is an adequate model for understanding the complex structure of atmospheric CO2 concentrations. For the future, we may consider using machine learning algorithms like LSTM or GRU to further optimize the prediction or to build a more robust model to improve the prediction.

Reference

1. Lan, X., Tans, P., & Thoning, K. W. (2025). Trends in globally-averaged CO_2 determined from NOAA Global Monitoring Laboratory measurements: Monthly Mauna Loa, Hawaii [Data set]. National Oceanic and Atmospheric Administration, Global Monitoring Laboratory. https://gml.noaa.gov/ccgg/trends/mlo.html